CHAPTER 72
ENGINE

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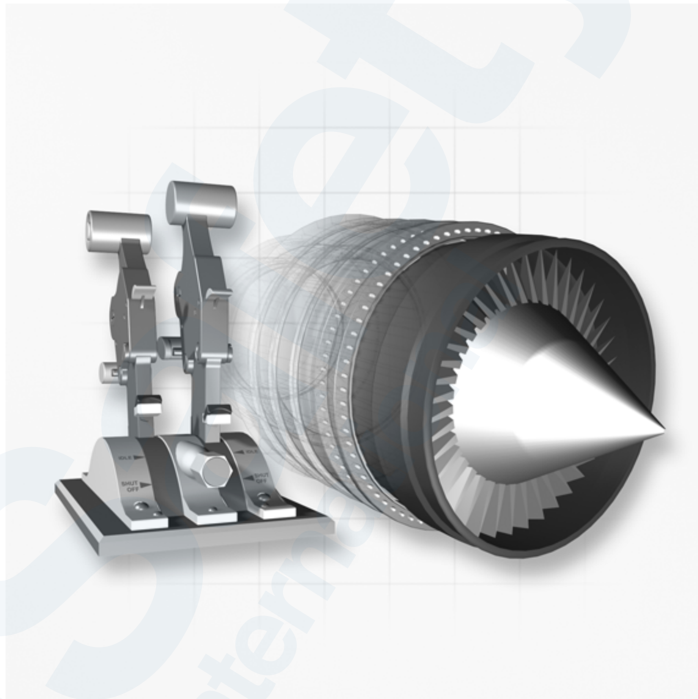
ILLUSTRATIONS

CHAPTER 72
ENGINE



72-00-00 INTRODUCTION

The Dash 8 series 400 aircraft is powered by two PW150A turboprop engines each able to produce 5071 SHP at Max takeoff.

The PW150A turboprop engine is a three spool, free turbine engine. The three spools are:

- Low Pressure spool
- High Pressure spool and
- Power Turbine spool.

The engine contains the following sub-systems:

- Reduction Gear Box (RGB) and Propeller
- Compressor Section
- Combustion Section
- Turbines.

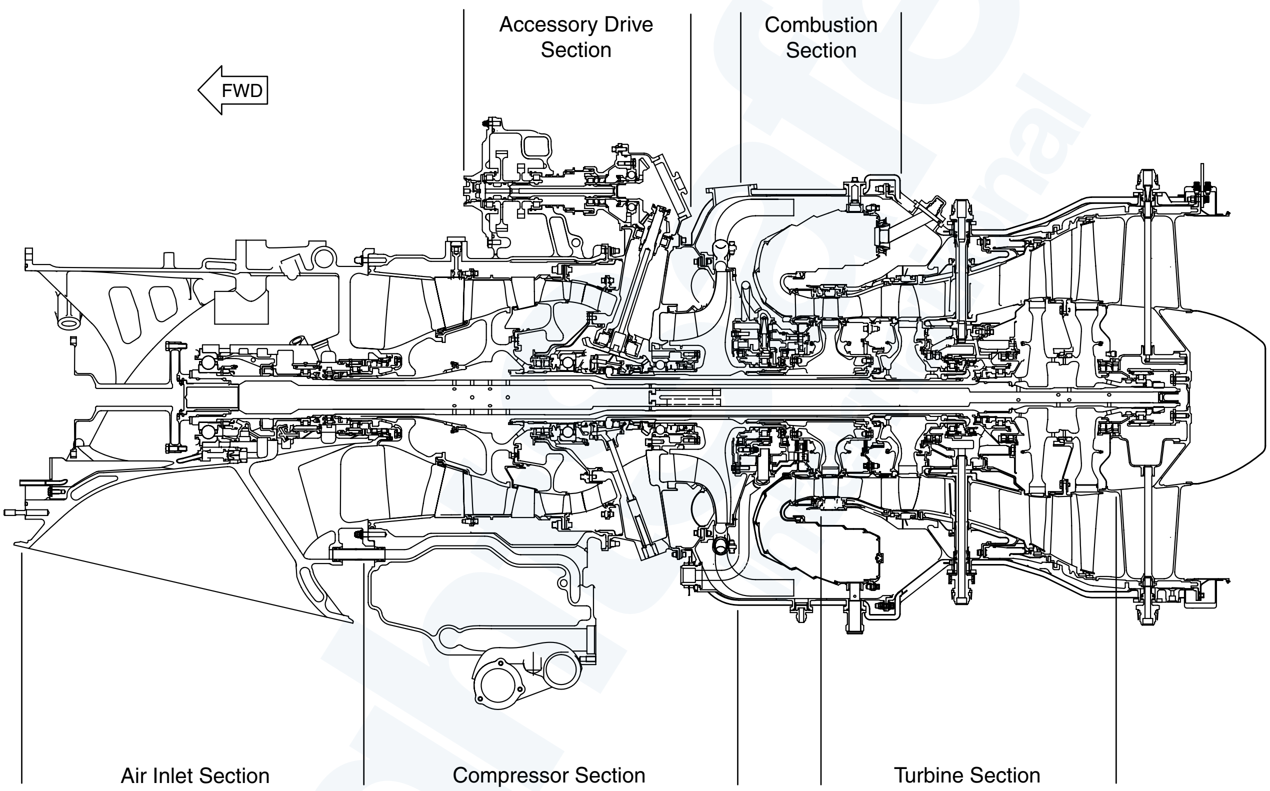


Figure 72-1. Engine Cutaway View 1

GENERAL

NOTES

Refer to Figure 72-1. Engine Cutaway View 1..

The Low Pressure (LP) spool rotates at a speed designated as N_L. The High Pressure speed designated as N_h.

The Power Turbine spool has a dual stage axial turbine that drives the Power Turbine shaft. This spool rotates at a speed designated as NPT, after reduction the speed is designated as N_p.

The HP and the Power-Turbine Spool rotate clockwise, and the LP spool rotates counterclockwise, when viewed from the rear.

The engine FMU is controlled by a dual channel FADEC. The propeller is controlled by a dual channel Propeller Electronic Controller (PEC).

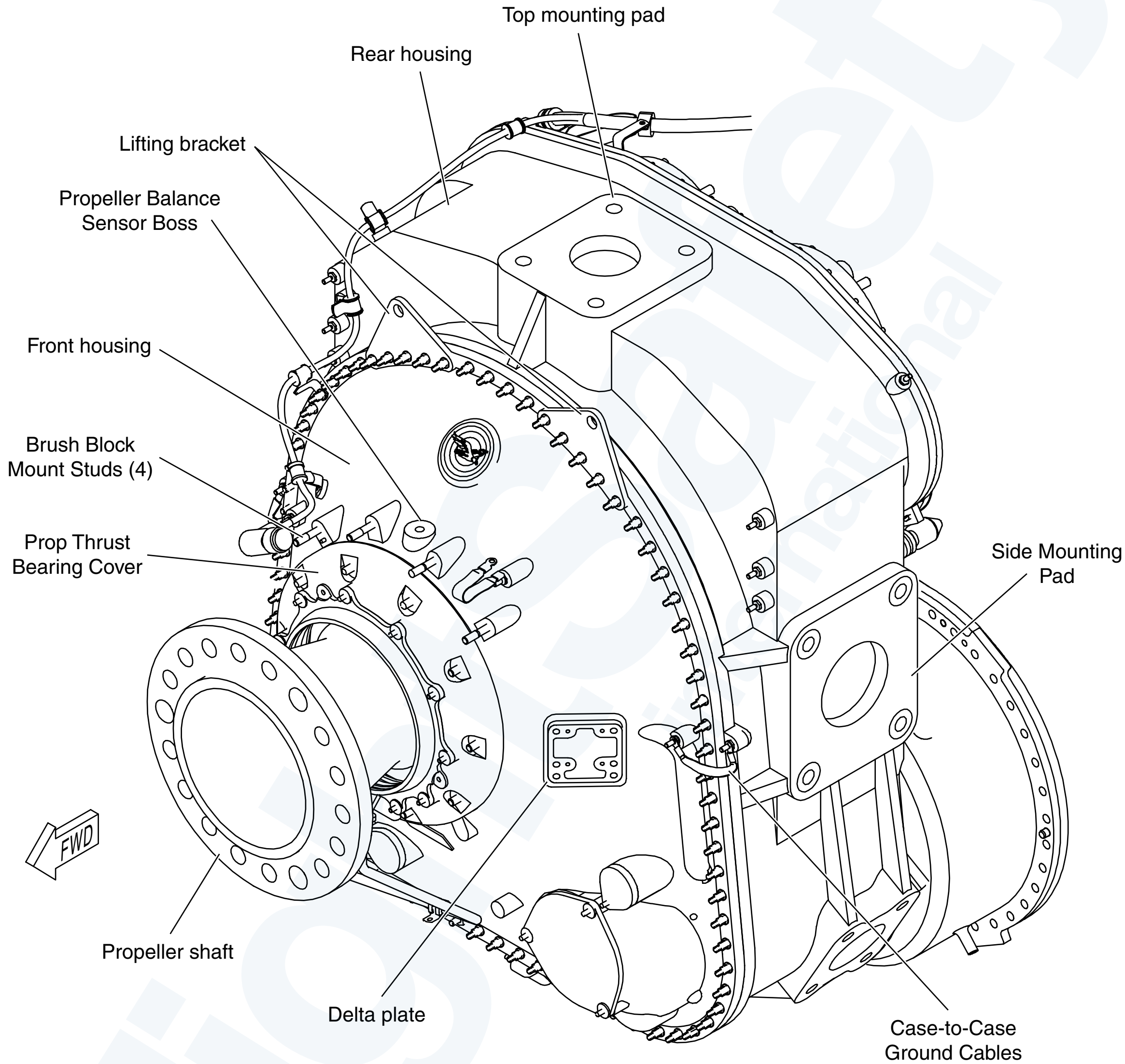


Figure 72-2. Reduction Gearbox (RGB)

72-10-00 REDUCTION GEAR BOX (RGB) AND PROPELLER SHAFT

GENERAL

Refer to Figure 72-2. Reduction Gearbox (RGB)..

The RGB has the following functions:

- Supports the propeller assembly
- Transmits the propeller thrust to aircraft structure
- Drives the following accessories:
  - The AC generator
  - The aircraft system hydraulic pump
  - The overspeed governor and pump
- Provides attachment for the three forward engine mounts
- Provides a housing for the RGB and AC chip detector.

The Aft face of the RGB has the following components:

- AC generator
- Hydraulic pump
- Overspeed-governor
- Pitch control unit (PCU)
- Alternate feathering pump and

  AC Generator chip detector.

The forward face of the RGB has the following components:

- Prop deice brush block
- Dual pulsed probe assembly (Magnetic Pick-up)
- RGB data plate.

NOTES

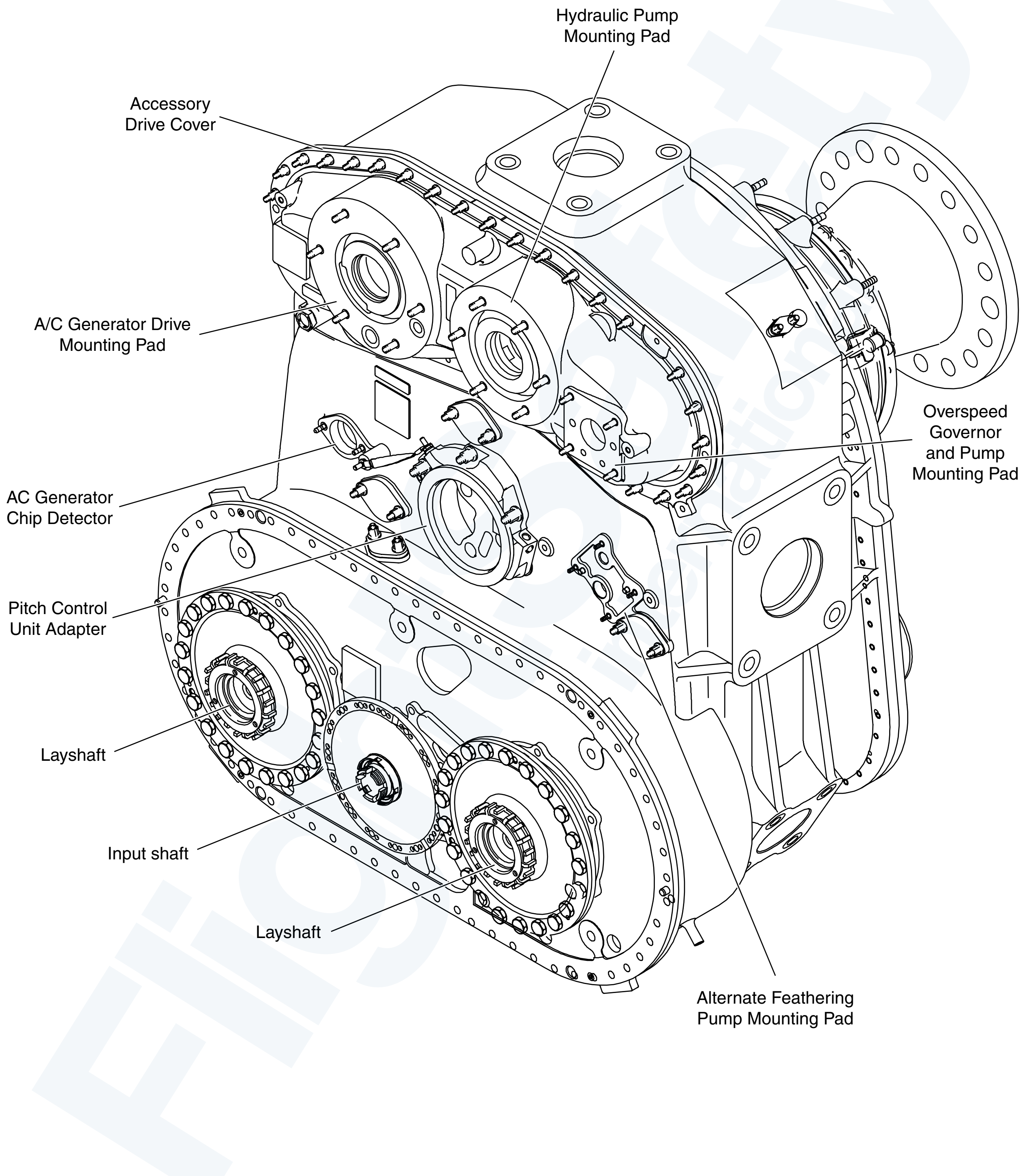


Figure 72-3. RGB - Rear Detail

SYSTEM DESCRIPTION

The RGB uses two stages of reduction to reduce the power turbine shaft RPM input and to increase the torque output of the propeller shaft.

COMPONENT DESCRIPTION

Reduction Gearbox

Refer to Figure 72-3. RGB - Rear Detail..

The RGB, on the inlet casing, include the following sub-components:

- Front housing
- Rear housing

  Diaphragm input-drive housing.

The RGB reduces the input speed from the power turbine to a speed range usable by the propeller.

There are two stages of reduction. The total reduction ratio is 17.16 to 1.

NOTES

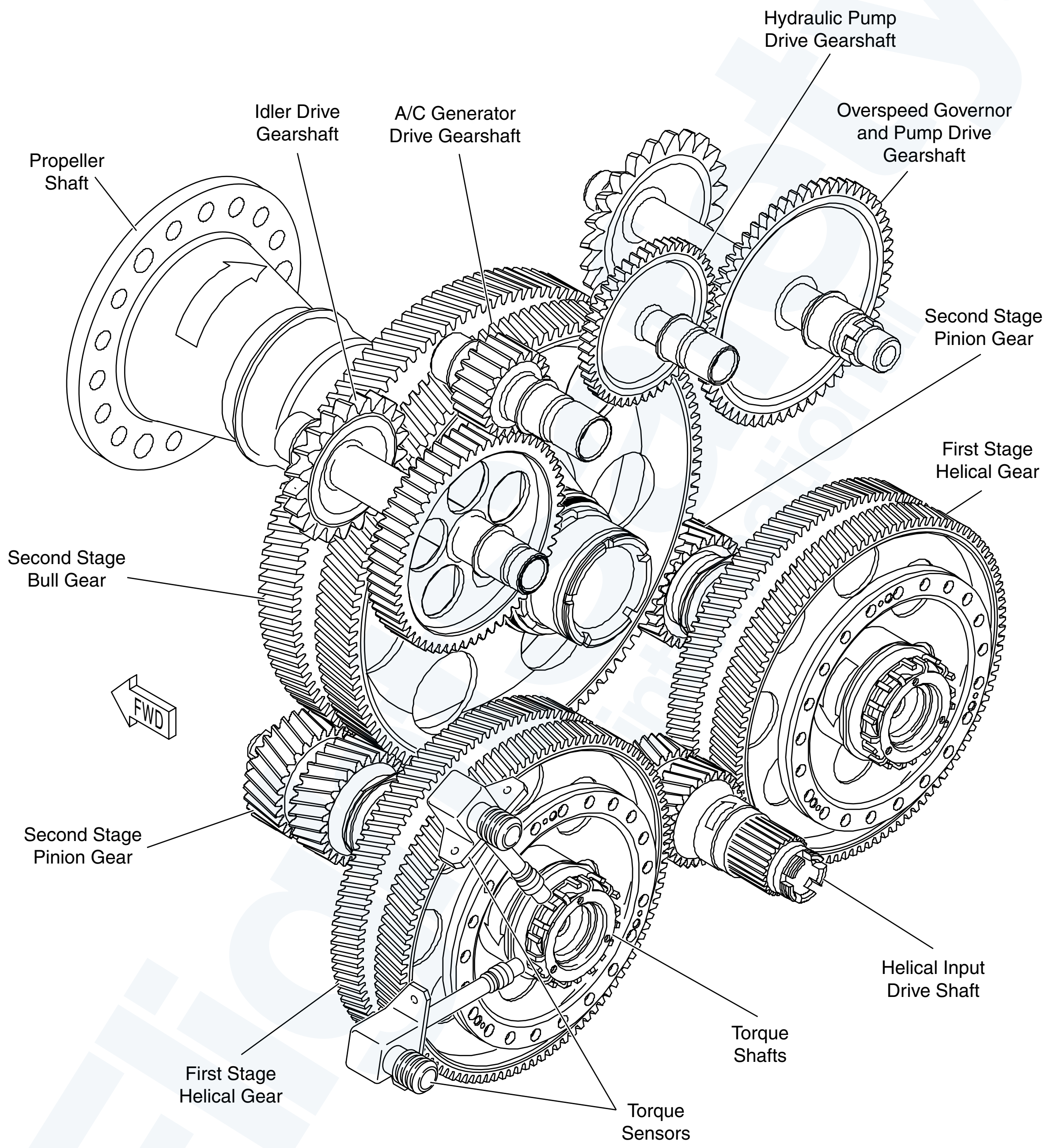


Figure 72-4. RGB Geartrain

OPERATION

Refer to:

- Figure 72-4. RGB Geartrain..
- Figure 72-5. RGB Geartrain.

The first stage of the reduction starts when the power turbine turns the helical input shaft clockwise. The helical input shaft will turn the two first-stage helical gear counter-clockwise. The first-stage helical gear is directly splined to the second stage pinion gear by the torque shaft.

The second stage of reduction starts when the two pinion gears turn the second stage bull gear. The second stage bull gear is splined to the Propeller shaft that rotates the Propeller clockwise.

The bullgear also turns the idler-drive spur gearshaft and the overspeed-governor gearshaft counterclockwise. The idler-drive gearshaft turns the alternator-drive gearshaft clockwise and the overspeed-governor gearshaft turns the Hydraulic pump drive spur gearshaft clockwise.

A chamber in the rear housing is supplied with engine oil. This auxiliary tank is always full of oil, even when the engine is not in operation. Oil from this tank lubricates the reduction and accessory gears and bearings through oil passages, oil jets and nozzles. The overspeed governor and the propeller control unit use this oil to operate the pitch-change mechanism of the propeller.

The electric feathering-pump mounting pad on the rear housing has oil ports that are connected to this auxiliary oil tank.

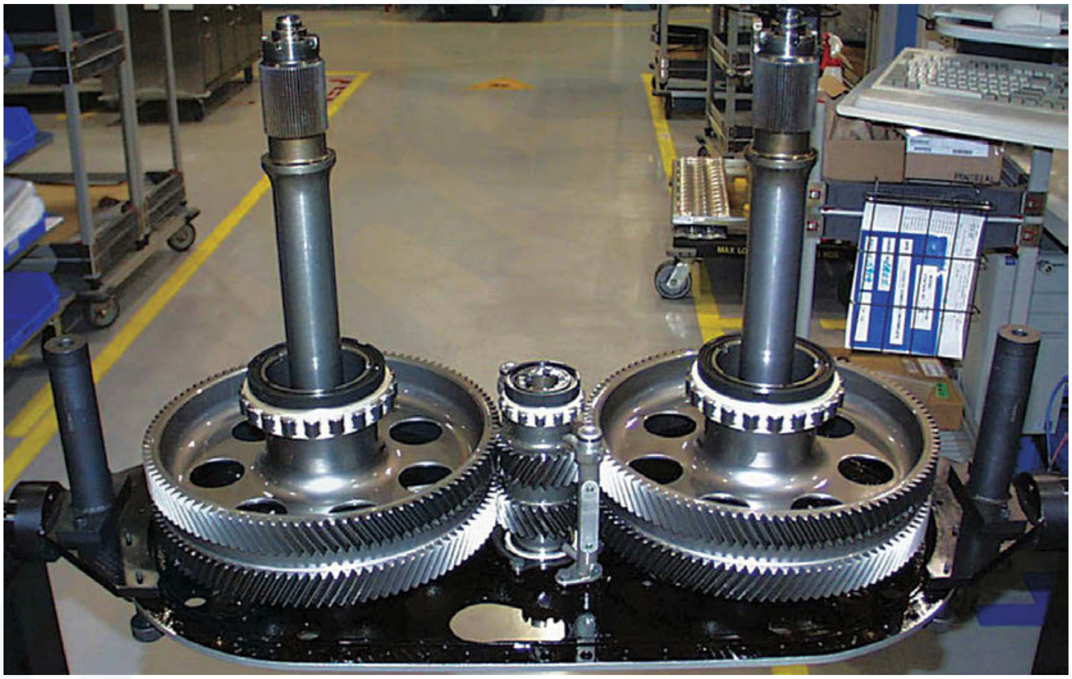


Figure 72-5. RGB Geartrain

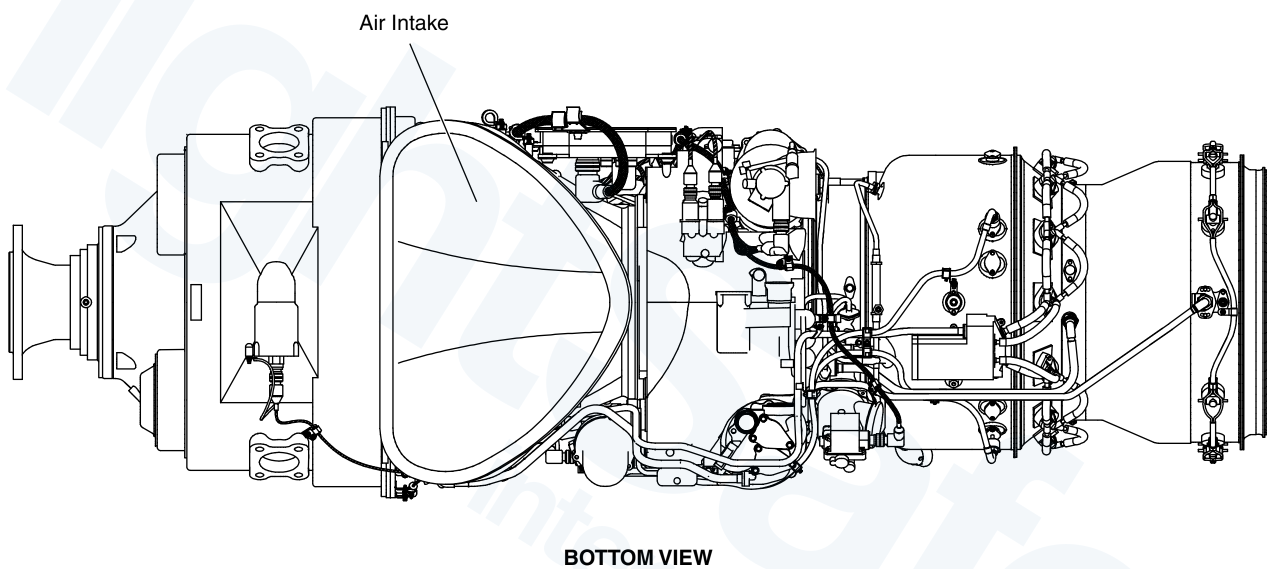


Figure 72-6. Engine Air Intake

72-20-00 INLET SECTION

GENERAL

The front inlet case directs air to the compressor section.

SYSTEM DESCRIPTION

The purpose of the front inlet case is to:

- Provide structural connection between the RGB and the turbomachinery
- Direct air flow between the intake housing and the first stage Integrated Blade Rotor (IBR)
- Support power turbine and low pressure shafts
- Support the flexible coupling shaft assembly that transmits power to the RGB
- Provide sealing for the No.1 bearing cavity
- Allow oil to pass to bearing cavities.

The FADEC is on the left side of the inlet case and the fuel flow meter is on the right side.

COMPONENT DESCRIPTION

Front Inlet Case

Refer to Figure 72-6. Engine Air Intake..

The air inlet is between the RGB and the Low Pressure (LP) compressor case. The inlet case provides air to the LP compressor.

The front inlet case is made of magnesium alloy. Hot oil flows through internal passages of the inlet case to prevent the formation of ice on the inlet lip. It houses the coupling shaft. It supplies installation for the following:

- FADEC
- Turbomachinery data plate
- Compressor wash port
- N_L Speed Sensor
- Torque and NPT sensors
- MOT Probe with ITT trim resistor
- T1.8 Probe.

CHROMATE SURFACE REPAIR OF MAGNESIUM

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

If the anti-corrosion coating on magnesium alloy is damaged it must be repaired as soon as possible.

There are two types of solution that may be used for this task as follows:

Chromate Conversion Solution or Chrome Pickle Solution.

They can be obtained ready mixed but tables are supplied in the Task Sheet detailing the solution mixtures.

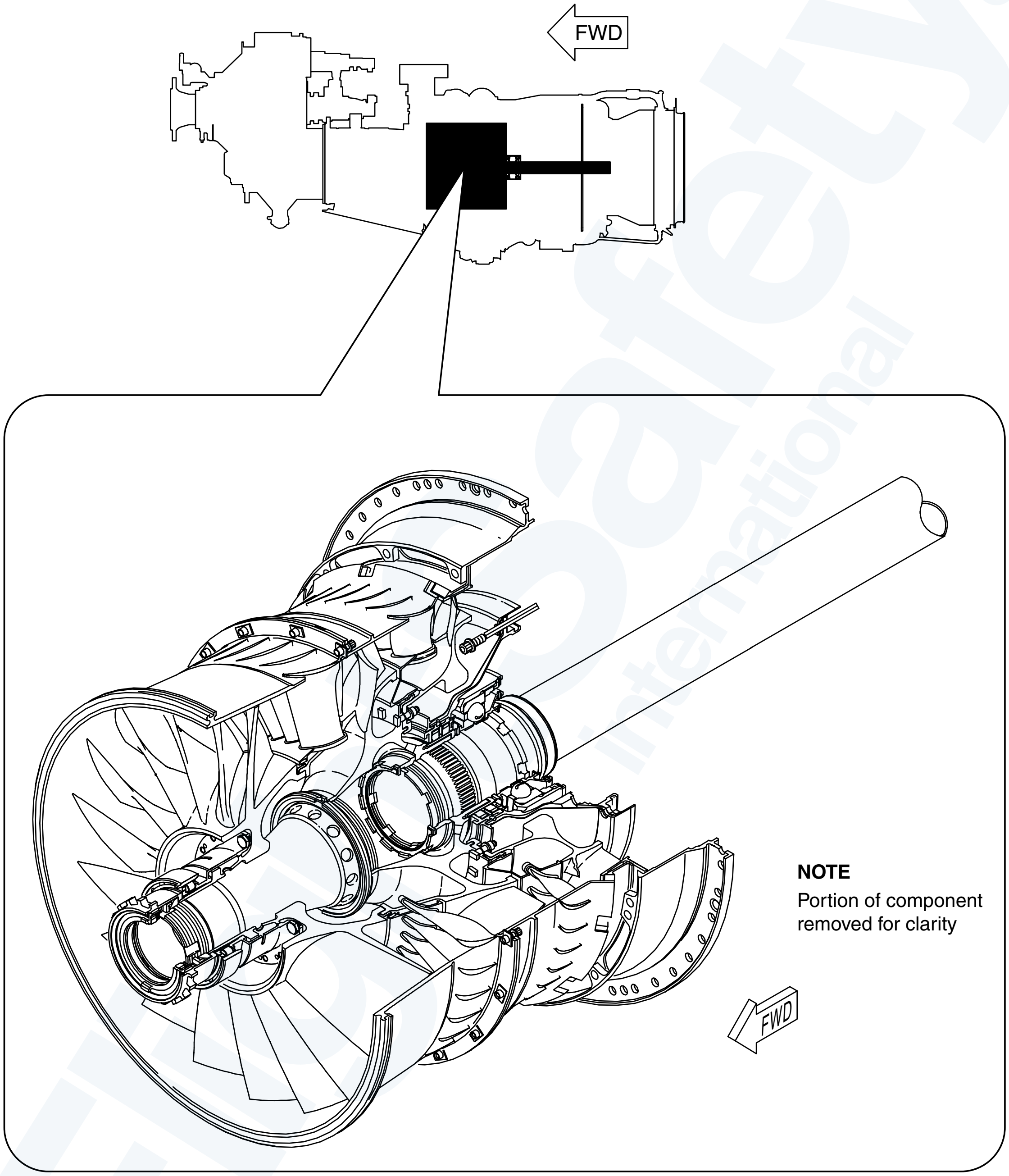


Figure 72-7. Compressor Section 1

72-30-00 COMPRESSOR SECTION

INTRODUCTION

The compressor section raises the pressure of the incoming air before passing it to the combustion chamber.

SYSTEM DESCRIPTION

Refer to:

- Figure 72-7. Compressor Section 1..
- Figure 72-8. Compressor Section 2..

The LP compressor case provides mounting for the following components:

- Two N_h speed sensors
- Oil level sight glass and filler neck
- Oil pressure regulating valve
- Main oil filter housing
- Ignition exciter box
- Fuel heater
- Turbomachinery chip detector
- Oil pump pack
- Deaerator
- Retimet breather (in AGB)
- P2.2 intercompressor bleed valve and adaptor.

The principal containment for the LP compressor blades is by the bolted vane and shroud arrangements.

The intercompressor case gives support for the following components:

- P2.7 check valve (for ECS)
- P2.7 handling bleed valve (HBV)
- Angle drive gearbox
- Rear engine mounts
- Drain valve and
- HP compressor.

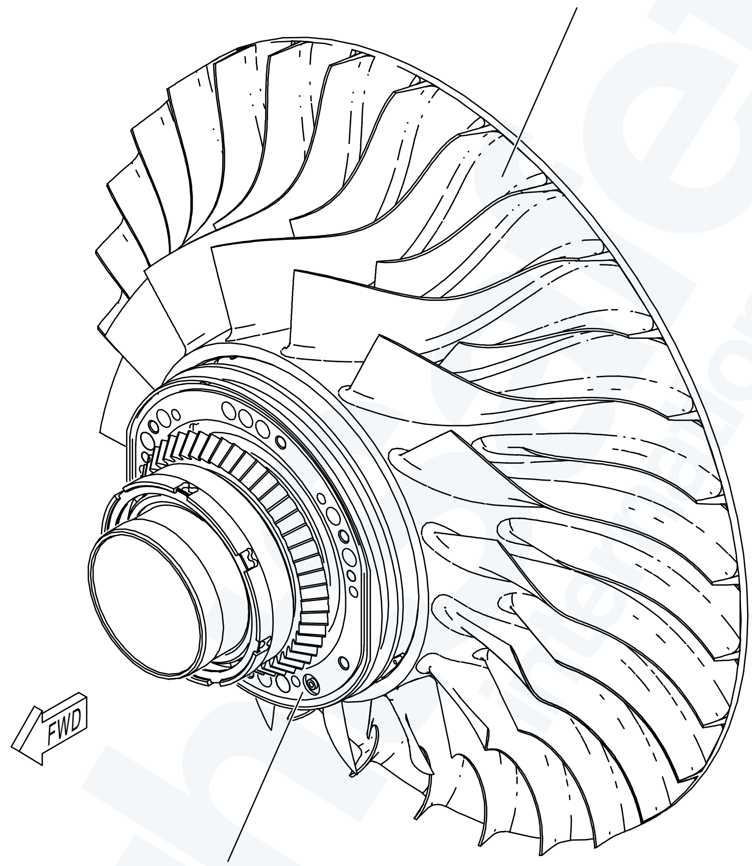


Figure 72-8. Compressor Section 2

COMPONENT DESCRIPTION

Low Pressure (LP) Compressor Case

The LP compressor case is between the inlet case and the intercompressor case.

This assembly is split in two halves to permit one piece compressor module assembly and removal.

The magnesium case offers some containment capability, through the erosion/thermal shield in the case.

An erosion shield is installed in the case to protect the magnesium from hot P2.2 air and debris.

The low pressure compressor case houses the low pressure compressor. The LP compressor case supports bearings No.2.5 and No.3.

The 1st stage vane assembly is a full vane ring cascade.

Low Pressure (LP) Compressor

The low pressure compressor is in the (LP) low pressure compressor case.

The low pressure compressor is an axial, three stage compressor and is driven by an independent axial turbine. The rotor components are three axial integrated blade rotors (IBR) (1st, 2nd, and 3rd stage) all made of titanium alloy. The IBRs are designed to be FOD resistant.

The LP compressor gives the first three stages of compression to the air mass going through the engine.

Inter-Compressor Case

The intercompressor case is between the (LP) low pressure compressor case and the combustion case.

It houses the centrifugal HP compressor and the accessory drive shaft (tower shaft).

The titanium intercompressor case holds the No.3 and No.4 bearing cavities that support the LP and HP compressors.

The impeller shroud is a double annulus for the P2.7, and P2.8 bleeds.

The intercompressor case forms a path that allows air to flow from the LP compressor to the HP compressor.

The compressor arrangement of three stages axial and a single stage centrifugal is prone to stall because at low N_h, the axial flow compressor is much more efficient than the centrifugal compressor. The air flows freely across the axial but not so freely across the centrifugal, leading to compressor stall and surge.

HP compressor

The high pressure (HP) compressor is in the intercompressor case, and is a centrifugal high pressure impeller driven by an independent axial turbine, on an integral shaft. The HP compressor gives the fourth and last stage of compression to the air mass going through the engine.

The bevel gear on the front of the compressor impeller meshes with the angle drive shaft, to provide the drive to the accessory gearbox.

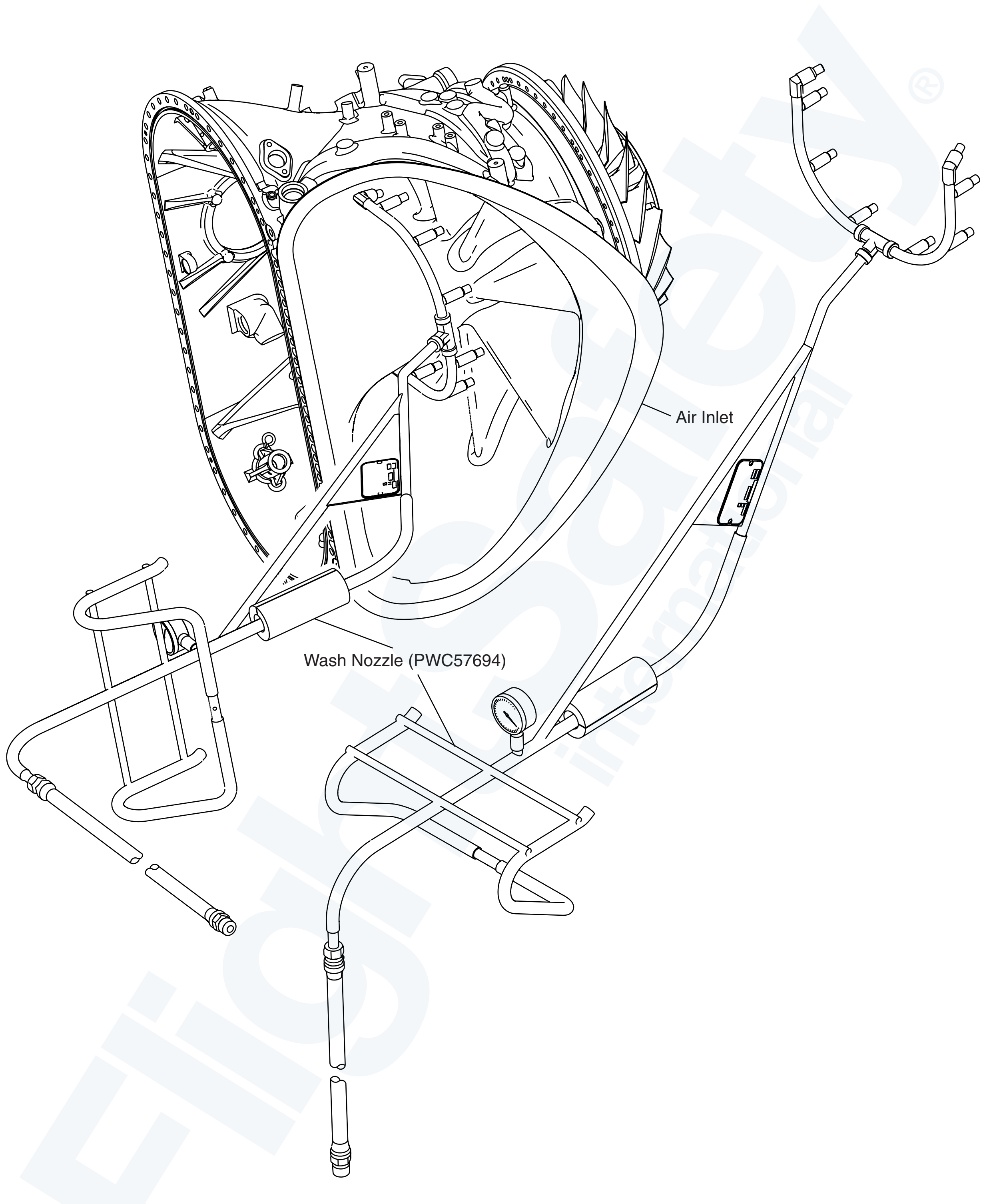


Figure 72-9. Wash Nozzle

OPERATION

Normal System Operation

The three stage axial compressor rotates counterclockwise (when viewed from the rear).

The 3rd stage stator assembly is a double row cascade. This double row cascade slows the air down to provide near zero swirl at the inlet to the Inter Compressor Case (ICC) duct.

Compressed air from the HP diffuser duct flows into the gas generator case around the combustion liner. The air flows into the dilution holes and through holes in the machined louvers. Air is used to assist in the fuel atomization, for cooling the liner skin, and for combustion.

The shaft is supported by the No.4 ball bearing and No.5 roller bearing.

Cabin bleed is provided by P3 and P2.7 air.

Gradual performance shifts of N_L, N_h, T_Q , W_F and ITT may be caused by a dirty compressor.

Sudden performance shifts of N_L, N_h, WF, and ITT may be caused by FOD or compressor rubbing.

COMPRESSOR AND TURBINE WASH

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

Refer to Figure 72-9. Wash Nozzle.

CAUTION

LET THE ENGINE COOL FOR A MINIMUM OF 40 MINUTES BEFORE WASHING THE COMPRESSOR. THIS WILL PREVENT DAMAGING ENGINE COMPONENTS.

A compressor and turbine wash removes salt, dirt and other baked-on material that collects in the gas path and can cause the performance of the engine to deteriorate. There are two types of washes that you can do to clean the compressor and turbines.

Desalination Wash

A desalination wash uses water or a water/methanol solution to remove salt and light deposits.

Performance Recovery Wash

A performance recovery wash uses cleaning chemicals in the wash solution to remove deposits that cannot be dissolved by a desalination wash.

Do the performance recovery wash when necessary to make sure that deposits do not build up on engine components.

A rinse wash is used after a performance recovery wash to clean the gas path.

There are two types of wash tooling. They are:

- PWC59053, Desalination Adapter - It attaches to the side of the engine inlet case
- PWC57694, Wash Nozzle Assembly - It installed into the engine inlet case. This can result in a more effective wash.

The use of a PWC57693 Wash Collector is recommended, but optional. This allows easy collection and disposal of the drained fluid.

ENGINE CONDITION TREND MONITORING (ECTM)

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

The EMU automatically records ECTM data during every take-off and stable cruise for every flight.

Engine trend monitoring is usually started a maximum of 100 flight hours from the time an engine is installed.

Download the data to a laptop computer which has the P&WC supplied Ground Based Software installed

Make an analysis of the ECTM data file (Refer to the P&WC ECTM Users Guide).

ECTM Data Retrieval Frequency:

With an EMU:

P&WC recommends that you download the ECTM data every 50 hours.

Without an EMU:

Automatic ECTM data collection is not available if the aircraft does not have an EMU. P&WC recommends that you record a data set each day or every eight flight hours, whichever comes first.

NOTE:

Record the parameters that follow:

- OAT
- P.ALT
- TQ
- NP
- NL
- NH
- ITT
- Wf

ECTM Data Analysis:

- To use ECTM satisfactorily, you must do a regular analysis of the data to find the condition of the engine. P&WC recommends that you analyze the ECTM data a maximum of five (5) days after you download it.
- ECTM can indicate a change in engine parameters. Compare the change to the performance margin for that engine. The performance margin was recorded during the power assurance check (PAC) done when the engine was initially installed.

72-40-00 COMBUSTION SECTION

INTRODUCTION

Refer to Figure 72-10. Combustion Section 1.

The combustion section burns the fuel air mixture and delivers the expanding gases to the turbine section.

NOTES

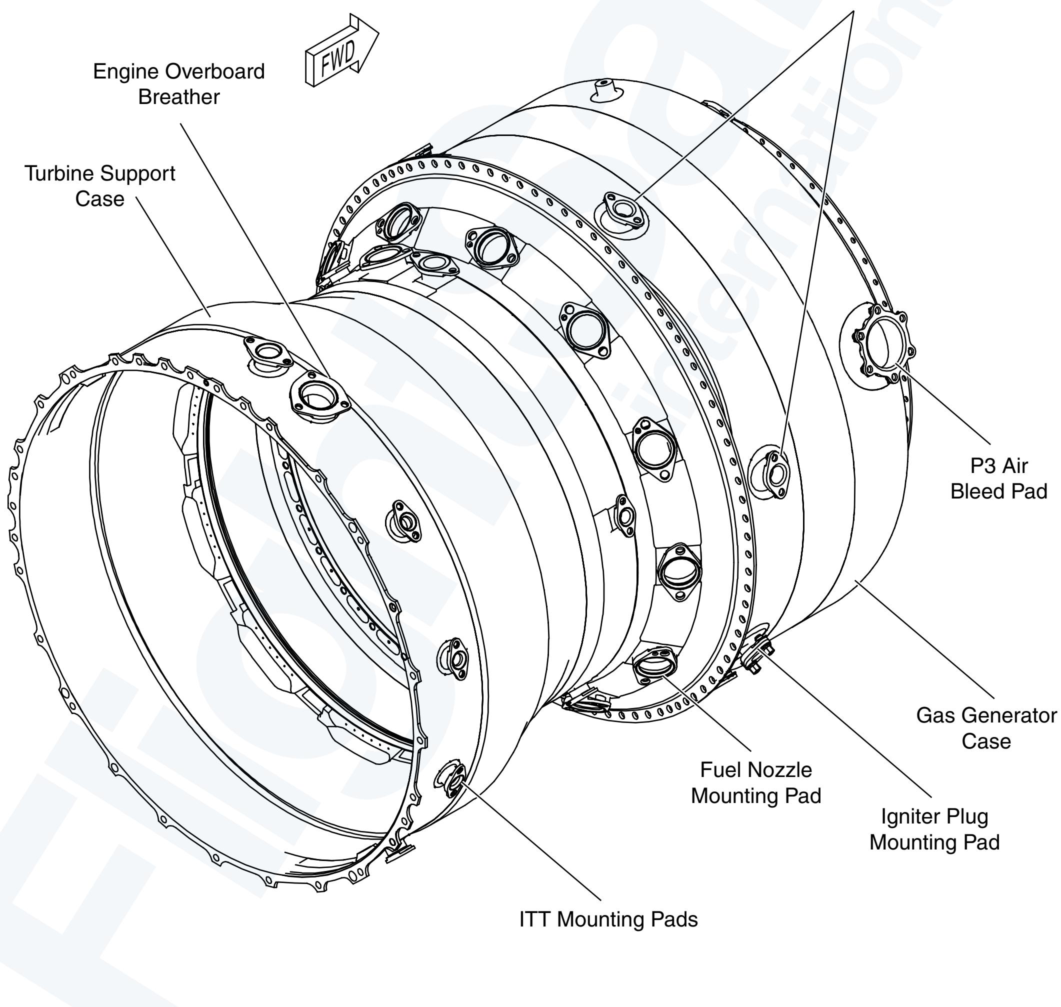


Figure 72-10. Combustion Section 1

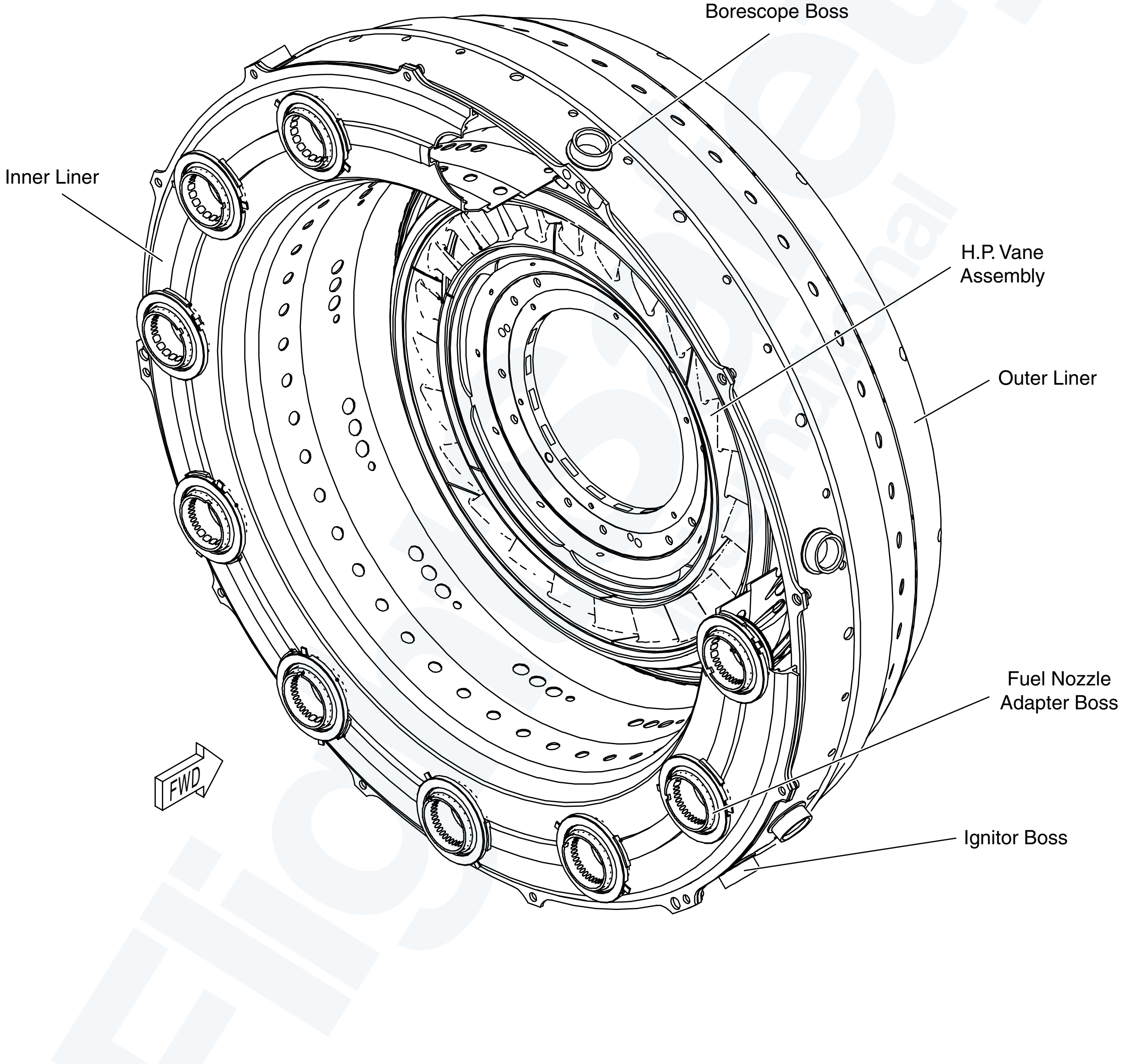


Figure 72-11. Combustion Section 2

SYSTEM DESCRIPTION

Refer to Figure 72-11. Combustion Section 2..

The gas generator case incorporates an air bleed pad through which P3 air is supplied to the Environmental Control System (ECS).

The inner surface comprises machined louvers. Two igniter plug bosses are provided on the gas generator case at the four and seven O'clock positions, with corresponding boss in the liner.

Combustor retention is provided by six combustor retention pins in the igniter plane.

The major components comprising the combustor are the:

- Outer liner
- Inner liner
- Small exit duct (SED)
- Adapter.

The large exit duct is double skinned, with impingement holes in the outer skin for effective cooling of the inner skin.

COMPONENT DESCRIPTION

Combustion Case

The combustion case is between the interturbine case and the Turbine section.

The combustion section provides an area for the combustion of the fuel/air mixture.

The Combustor

The annular reverse flow combustion chamber is contained in the gas generator case. It is at the rear of the HP compressor and the fish tail diffuser ducts.

The combustor burns the mixture of fuel and air, and delivers the resulting gases to the turbines.

OPERATION

Twelve hybrid fuel nozzles each containing a primary air blast and a secondary air blast with combustor air swirler, are incorporated. Fuel is added to the compressed air in the combustion chamber.

Two igniter plugs are provided on the gas generator case with corresponding bosses in the combustion chamber liner. The igniters are used for starting but are not required once the fuel/air mixture is lit.

Hot combustion gases flow forward in a three dimensional torroid and are directed rear toward the HP vane by the combustor outer liner and the small exit duct.

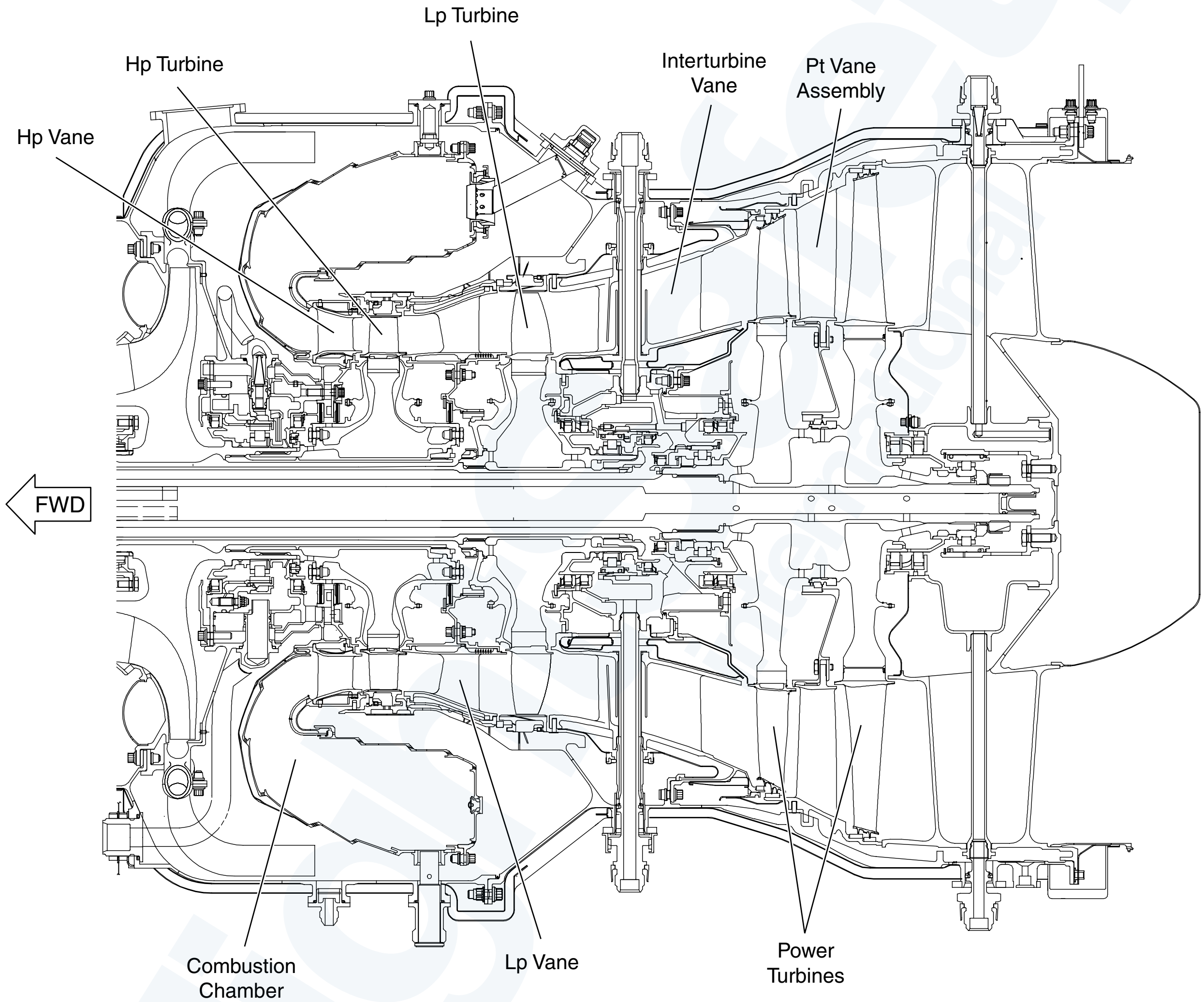


Figure 72-12. Turbine Section 1

72-50-00 TURBINE SECTION

GENERAL

Refer to:

- Figure 72-12. Turbine Section 1..
- Figure 72-13. Turbine Section 2..
- Figure 72-14. Turbine Section 3..
- Figure 72-15. Turbine Section 4..

The turbines extract kinetic energy from the expanding gases as the gases flow from the combustor. The turbines then convert this energy into shaft horsepower to drive the compressors, propeller and the engine accessories.

The hot section of the engine has the following three turbine stages:

- Low Pressure (LP) turbine
- High Pressure (HP) turbine
- Power Turbines (PT).

NOTES

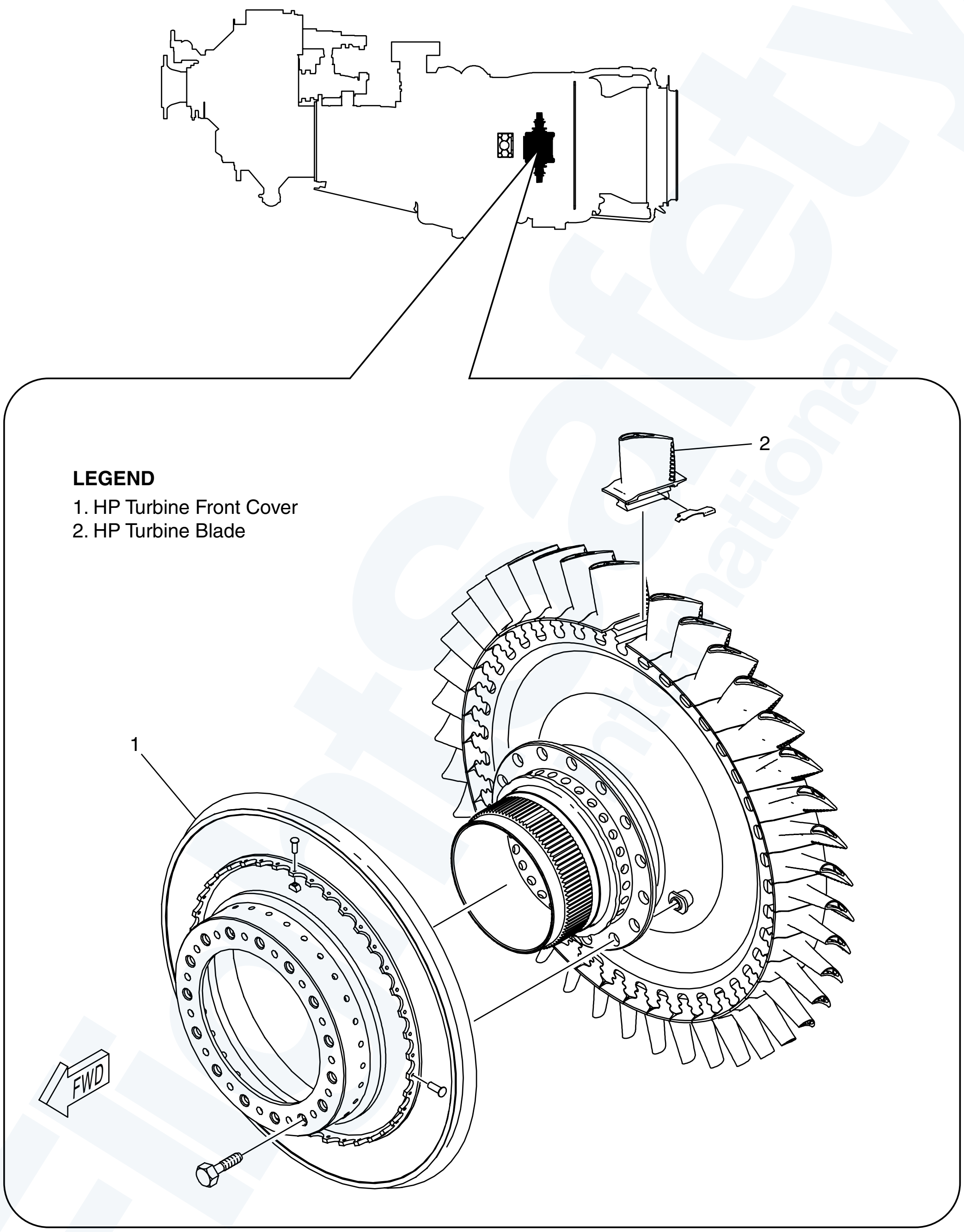


Figure 72-13. Turbine Section 2

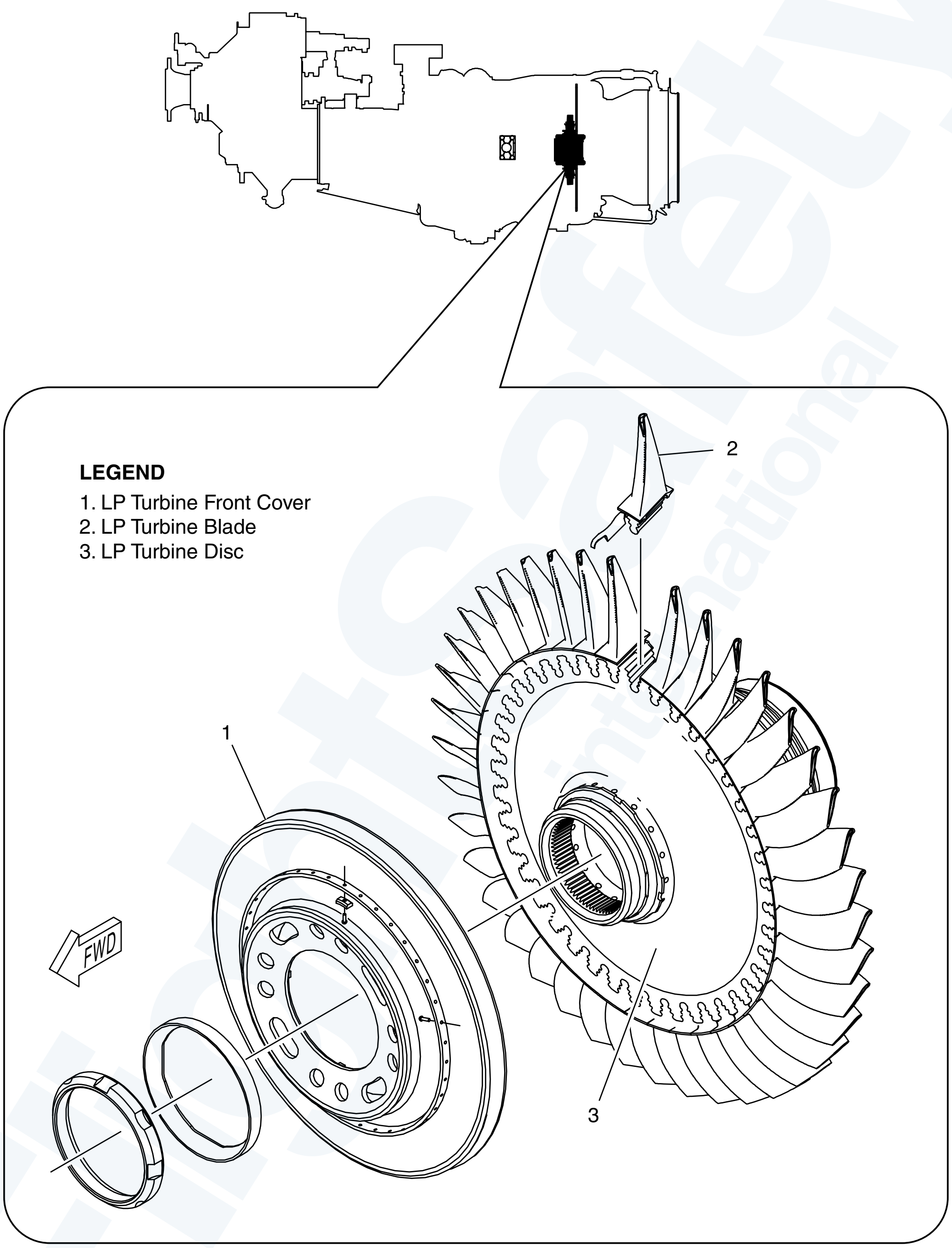


Figure 72-14. Turbine Section 3

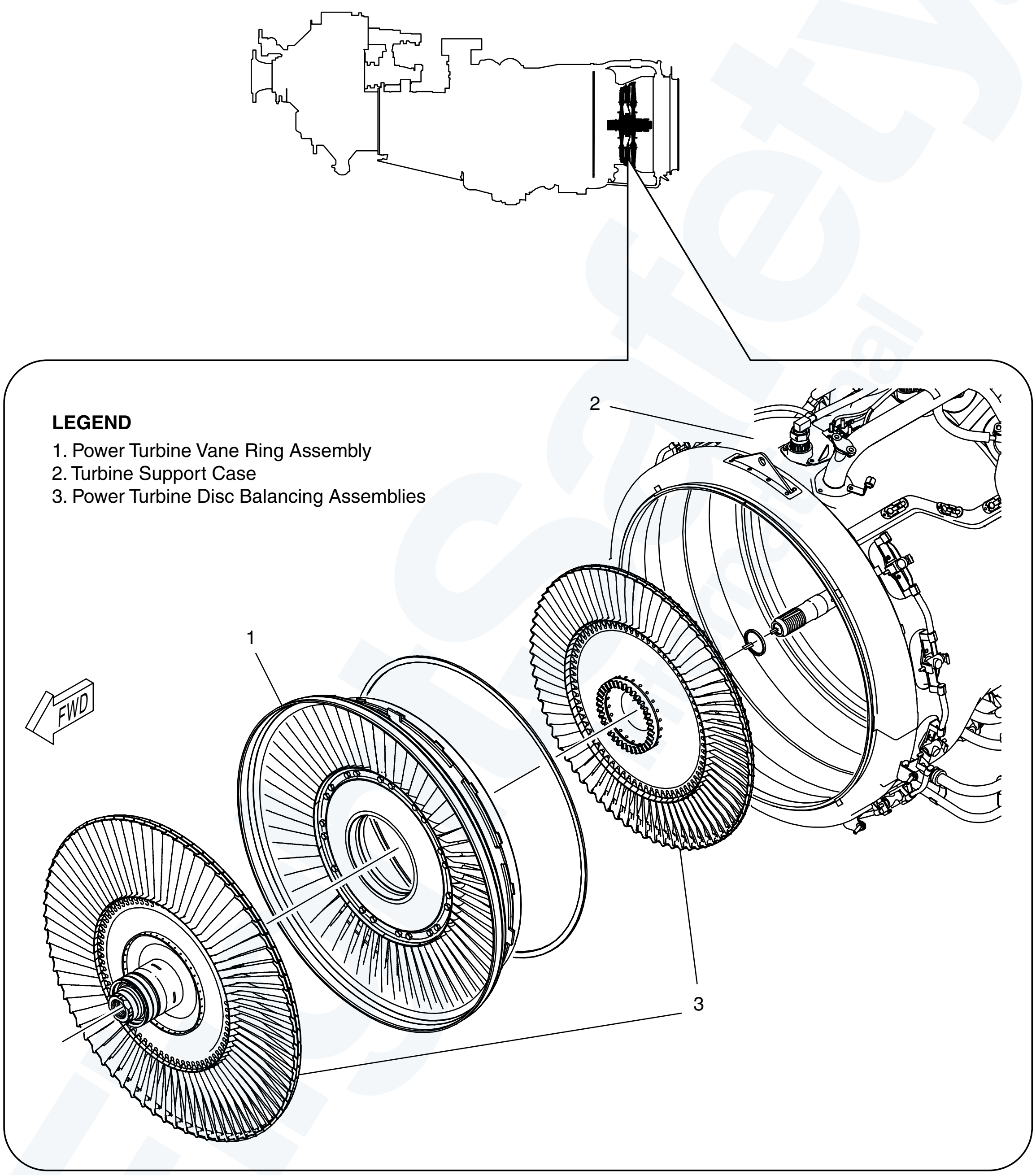


Figure 72-15. Turbine Section 4

SYSTEM DESCRIPTION

A ring of stator vanes is installed in front of its associated turbine, to direct the hot gases to the turbines and change static pressure into velocity.

The LP and HP turbines are installed in the rear of the gas generator case, and the power turbines are installed in the turbine support case.

The hot section also has the following vane assemblies:

- HP vane assembly
- LP vane assembly
- Inter-turbine vane assembly and
- Power turbine vane assembly.

The central PT shaft is supported by the No.1 (ball), No.2 (roller), No.6.5 (roller) and No.7 (roller) bearings.

The intermediate LP turbine shaft is supported by the No.2.5 (roller), No.3 (ball) and No.6 (roller) bearings.

The HP turbine shaft, that is integral with the impeller, is supported by the No.4 (ball) and No.5 (roller) bearings.

The hot section of the engine comprises components downstream of the gas generator.

COMPONENT DESCRIPTION

HP Vane

The stator vanes ring is installed in front of the HP turbine section in the gas generator case and air cooled using P3 air.

Support comes from the Small Exit Duct (SED) and the inner support housing.

The purpose of the HP vane assembly is to direct the hot gases to the HP turbine and change static pressure into velocity.

HP Turbine

The HP turbines are installed in the rear of the gas generator case.

The HP turbine consists of a nickel alloy disc featuring 41 air cooled blades. The blades are secured by fir-tree serrations and two covers that retain the blades axially and minimize cooling air leakage through the fir-tree of the blades.

The cooling air is provided through showerhead, tip and platform cooling holes with trailing edge ejection. The front cover also increases the air pressure delivered. The cooling air comes from the Tangential Outboard Injector (TOBI). Cooling air is used for the following:

- Cool the HP blades
- Ventilate the HP disc bore
- Purge the downstream side of the rear cover
- Impinge upon the LP vane inner drum.

The HP turbine extracts energy from the hot gases to turn the HP Compressor and the accessory gearbox.

LP Turbine Vane Assembly

Located in the Gas generator Case between the HP turbine and the LP turbine.

The purpose of the LP vane assembly is to direct the hot gases to the LP turbine and change static pressure into velocity.

LP Turbine Assembly

The LP turbines are installed in the rear of the gas generator case.

The LP Turbine disk is made of nickel alloy featuring 41 air-cooled blades. The base material for the blades is a single crystal nickel alloy. The blades are secured by fir-tree serrations. The blades use trailing edge ejection for the cooling air. The disc is straddle mounted on the LP shaft.

LEGEND
① For 20 seconds
② For 120 seconds
③ For 600 seconds

OPERATING LIMITS

Operating Condition	Power	Torque	Performance			EGT	NH	NP	Oil Pressure	Oil Temp.
	SHP OAT	Lb ft (%)	ESHP	Jet Thrust	Max. SFC Lb/ESHP/Hr	°C	RPM (%)	RPM (%)	PSID	°C
Take-off	5071 37.4 °C	26113 100%	5492	843	.433	880	31150 (100)	1020 (100)	61-72	0-107
Normal Take-off	4580 37.4 °C	26113 100%	4963	767	.433	880	31150 (100)	1020 (100)	61-72	0-107
Max. Continuous Enroute Emergency	5071 37.4 °C	26113 100%	5492	843	.433	880	31150 (100)	1020 (100)	61-72	0-107
Max. Climb	4058 30.5 °C	26113	4058	688	.460			900		
Max. Cruise	3947 25.8 °C	26113	3947	671	.464			850		
Min. Idle							(64.2)		44 to 75 Min.	-40-125
Starting						920 ①			100Max>0°C 165Max<0°C	-40 Min.
Transient		35252 135% ① / 27680 106% ③				920 ①	31525 (101.2)	1071-1173 (105-115)	72-80 ② 80-100 ①	125 ①
Max. Reverse	1500									

Figure 72-16. Operating Limitations

The purpose of the LP turbine is to extract energy from the expanding gases to turn the LP Compressor.

Turbine Case

The Turbine case is after the gas generator case.

The turbine case supports the power turbines.

Each disc is installed and removed separately. This eases assembly and disassembly of the power turbine section.

The nozzles are on the turbine support case and protrude into the combustion chamber liner from the rear.

Inter Turbine Vane (ITV) Assembly

The ITV is attached to the TSC at a flange.

The purpose of the LP vane assembly is to direct the hot gases to the LP turbine and change static pressure into velocity.

In the event of a LP shaft failure, the ITV is designed to contain the LP rotor axially.

The ITV performs the following functions:

- Acts as the inter turbine duct
- Acts as the PT1 vane
- Provides support to the No. 6 and No. 6.5 bearing housings.

Four borescope ports are part of the ITV for gas path component inspection.

Power Turbines

The power turbines extract energy from the hot gases to turn the propeller through the RGB. There is a vane assembly between the rotors. The rotor assemblies consist of 72 shrouded blades held to the discs by fir-tree base and a rivet. The 1st stage blades are made of Nickel alloy 2nd stage blades made of Inconel alloy. The Power Turbines discs are made of Nickel alloy.

The purpose of the power turbines is to extract energy from the hot gases to turn the propeller through a reduction gearbox.

Exhaust Flange Cover

The covers are installed on the exhaust adaptor flange, forming two halves on the exhaust adaptor flange.

The cover forms a seal at the connection of the exhaust adaptor flange and the exhaust nozzle.

OPERATION

Normal System Operation

Hot expanding gases leaving the combustion chamber are directed towards the HP turbine blades by the HP turbine ring. After the HP turbine the hot gases are directed towards the LP turbine blades by the LP turbine ring.

The gases then travel across the PT vane rings and impinge on the PT turbine blades.

All three turbines turn at independent speeds. At engine start, the starter/generator turns only the HP compressor and HP turbine.

The HP turbine rotates clockwise to a maximum speed of 31,150 rpm (100%). The LP turbine rotates counterclockwise to a maximum speed of 27,000 rpm (100%). The Power turbine rotates clockwise to a maximum speed of 17,501 rpm (100%).

The LP and HP turbines turn the compressors through their own respective shafts. The power turbines turn the propeller through the power turbine shaft and the reduction gearbox.

Concentric shafts connect the two-stage power turbine to the gearbox and the single-stage LP and HP turbines to the compressors.

Limitations

Refer to Figure 72-16. Operating Limitations..

Borescope Inspections

There are various borescope inspection access points throughout the turbomachinery as follows:

- Inlet case - 1st stage LP rotor
- LP compressor case - 2nd and 3rd stage LP rotor
- Combustor support pins - HP impeller, combustion chamber liner, SED and HP stator and shroud segments. See NOTE 1
- Rear access cover of Starter/Gen drive - HP impeller vanes
- Turbine support case - LP turbine blades, Interturbine vane ring and 1st stage PT blades. See NOTE 1
- AGB breather tube cover - 2nd stage PT and Exhaust duct
- 10 o'clock position on Gas Generator case - No.5 bearing oil pressure and scavenge tubes
- P3 bleed air adapter - No.5 bearing scavenge tube leakage into bottom of gas generator case.

NOTE 1

Special Detailed Inspections (SDIs) that together constitute a Hot Section Inspection and utilize some of these access ports, are carried out by trained, skilled operators at the specified intervals (4000 Engine Hours repeat 1500 EH).

The only internal visual inspection that can be performed on the engine at field level is boroscope inspection.

CAUTION

YOU MUST BE VERY CAREFUL WHEN YOU USE THE BORESCOPE IT CAN BE EASILY DAMAGED BY HEAT, SHOCK, TWISTING AND PINCHING. TO PREVENT DAMAGE TO EQUIPMENT OBSERVE THE FOLLOWING ITEMS:

- ◦ DO NOT PUT THE BORESCOPE INTO LIQUID
- ◦ ENGINE TEMPERATURE MUST BE LESS THAN 66°C (150°F)
- ◦ BEFORE YOU TURN THE HP IMPELLER, REMOVE THE FIBERSCOPE FROM THE ICC
- ◦ THE PROTECTIVE RING AND SIDE-VIEWING ADAPTER CAN FALL INTO THE ENGINE IF NOT INSTALLED CORRECTLY
- ◦ THE PROTECTIVE RING AND SIDE-VIEWING ADAPTER CAN DAMAGE THE DISTAL END IF IT IS OVER TIGHTENED
- ◦ TO PREVENT DAMAGE TO THE OPTIC FIBERS TURN THE FIBERSCOPE WITH THE HAND NEAR THE POINT OF ENTRY NOT BY THE EYEPIECE. OVER TWISTING THE FIBERSCOPE
- ◦ TO PREVENT DAMAGE TO THE COMPRESSOR CAREFULLY TURN THE COMPRESSOR WITH A WOODEN OR PLASTIC DOWEL
- ◦ UNWANTED MATERIAL CAN CAUSE A BLOCKAGE OF THE NOZZLE.

ENGINE SHIPPING METHODS

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

This task describes two methods of shipping for engines as follows:

- Engine in shipping container.
- Engine shipped in a transportation stand.
- If shipping a QEC must be shipped in a transportation stand.

72-00-00 SPECIAL TOOLS & TEST EQUIPMENT

- Local Manufacture Transportation Stand
- PWC34910-101 Borescope assembly
- PWC34960-201 Camera (optional) (Consists of an Olympus OM-2 Camera incorporating a 50 mm f:1.8 lens and a 1-9 focusing screen)
- PWC34912 Accessory Kit
- PWC34913 Fixture, Holding
- PWC37711 Borescope Kit
- PWC57233 Puller
- PWC55829 Puller
- PWC57320 Guide Tube
- PWC55607 Puller
- PWC57533 Guide tube, Borescope (used in the second procedure only)
- PWC57466 Eddy Current Inspection Kit
- Uniwest 96400 Eddy Current Unit (ECU) US-454
- Uniwest FET3218 Eddy Current Probe for Concave Side of Airfoil
- Uniwest FET3219 Eddy Current Probe for Convex Side of Airfoil
- Uniwest 94032 Eddy Current Cable Assembly
- PWC59005 Eddy Current Calibration Standard
- PWC59053 Desalination Adapter
- PWC57693 Wash Collector
- PWC57694 Wash Nozzle Assembly
- PWC32677-100 Wash Cart
- Unitek 250DP or equivalent Micro Welding Equipment
- Commercially Available Keensert driving tool, Kee TD624L
- Commercially Available Helical coil insert extraction/installation tool
- Commercially Available Helical coil tang removal tool
- Commercially Available Helical coil thread plug gage and bottoming tap (standard thread size)
- Commercially Available Helical coil thread plug gage and bottoming tap (oversize thread size)
- PWC57486 Spreader
- PWC57487 Spreader
- PWC64325 Drift
- PWC57358 Gauging Plug
- PWC57183 Guide

- PWC57183 Plug tap
- PWC57183 Bottoming tap
- PWC57183 Puller
- PWC57131 Drift
- R1113W Wrench
- R212D Lockring Drive Tool
- PWC55971 Engine Stand
- GSB7100021 Nacelle-Mounted Engine Hoist
- PWC55971 Engine Stand
- GSB7100021 Nacelle-Mounted Engine Hoist

72-00-00 MAINTENANCE PRACTICES

Refer to the Bombardier AMM PSM 1-84-2 for details on these maintenance procedures:

- AMM 72-00-00-290-801: Borescope Inspection - General.
- AMM 72-00-00-290-802: Borescope Inspection of the 1st. Stage LP Compressor Rotor and Stator.
- AMM 72-00-00-290-803: Borescope Inspection of the 2nd and 3rd Stage LP Compressor Rotors.
- AMM 72-00-00-290-804: Borescope Inspection of the HP Impeller.
- AMM 72-00-00-290-809: Borescope Inspection of the Compressor Inner Support and the Intercompressor Case Struts.
- AMM 72-00-00-290-805: Borescope Inspection of the Combustion Chamber Liner Assembly, Small Exit Duct, HP Turbine Vane Segments, HP Shroud Segments, and HP Turbine Blades.
- AMM 72-00-00-290-808: Borescope Inspection of the Gas Generator Case.
- AMM 72-00-00-290-806: Borescope Inspection of the Low Pressure Turbine Blades, Low Pressure Vane Segment, Low Pressure Shroud Segment, Interturbine Vane Struts, and First-Stage Power Turbine Blades.
- AMM 72-00-00-290-807: Borescope Inspection of the Second–Stage Power Turbine Blades and Exhaust Duct.
- AMM 72-00-00-890-803: Borescope Inspection of the 2nd and 3rd Stage LP Compressor Rotors.
- AMM 72-00-00-160-801: External Wash - Using Water.
- AMM 72-00-00-160-803: Compressor and Turbine Wash.
- AMM 72-10-00-290-801: Borescope Inspection of the Reduction Gear Box (RGB) First–Stage Helical and Input Shaft Gear.

- AMM 72-10-00-290-802: Borescope Inspection of the Reduction Gear Box (RGB) Second-Stage Bull Gear, Helical Gear and Input Layshaft Pinions.
- AMM 72-40-00-280-801: Special Detailed Inspection (Borescope) of the Combustion Chamber Liner Components (Do this task in conjunction with #725000-202) (MRB#724000-201).